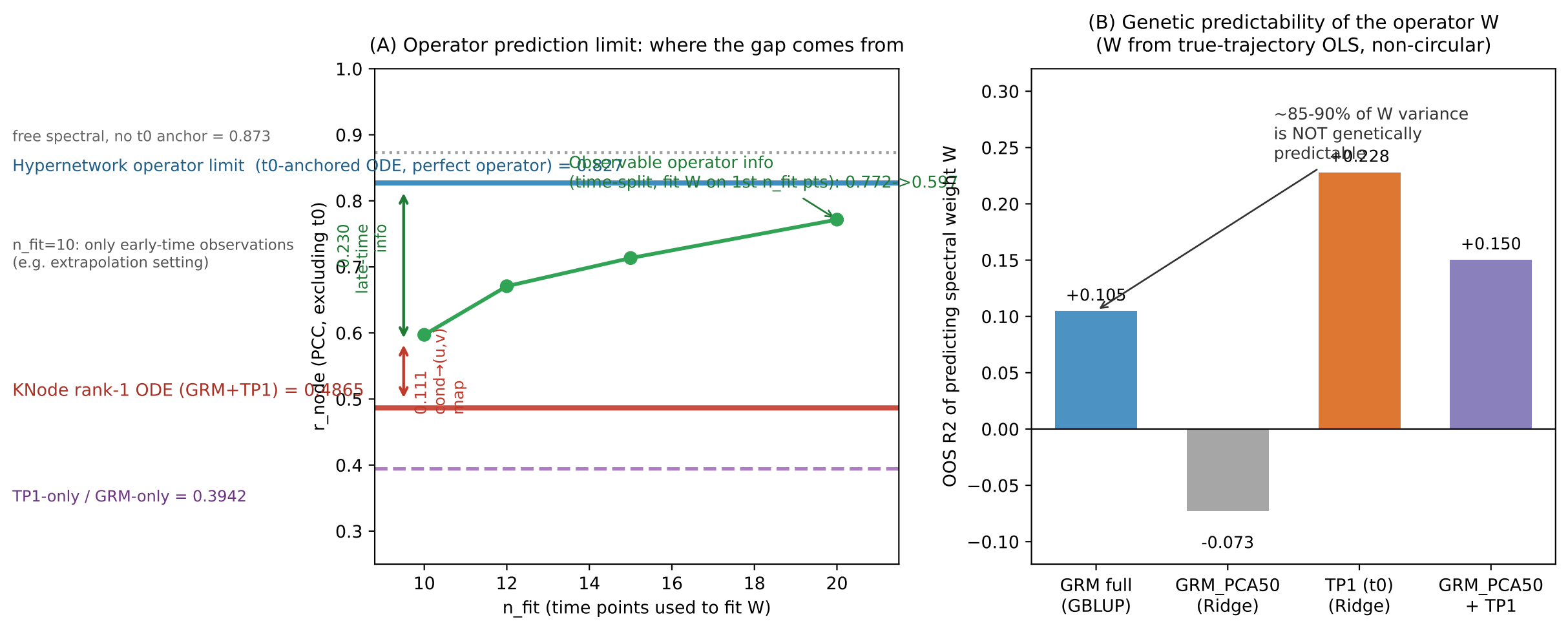
(A) Operator prediction limit: where the gap comes from
free spectral, no t0 anchor = 0.873
Hypernetwork operator limit (t0-anchored ODE, perfect operator) = 0.827
n_fit=10: only early-time observations
(e.g. extrapolation setting)
Observable operator info
(time-split, fit W on 1st n_fit pts): 0.772->0.59
0.230
late-time
info
0.111
cond→(u,v)
map
KNode rank-1 ODE (GRM+TP1) = 0.4865
TP1-only / GRM-only = 0.3942
r_node (PCC, excluding t0)
n_fit (time points used to fit W)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
10
12
14
16
18
20
(B) Genetic predictability of the operator W
(W from true-trajectory OLS, non-circular)
~85-90% of W variance
is NOT genetically
predictable
+0.105
-0.073
+0.228
+0.150
OOS R2 of predicting spectral weight W
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
GRM full
(GBLUP)
GRM_PCA50
(Ridge)
TP1 (t0)
(Ridge)
GRM_PCA50
+ TP1